# Exercises and Complete Solutions

Vector Spaces, Bases, Rank and Projections · v1.0

Work each problem before reading its solution. Vectors are real column vectors unless stated otherwise; semicolons separate matrix rows. Numerical checks allow floating-point tolerance. These exercises extend the owner-approved chapter and notebook and await review as new production assets.

## Level A · Subspaces and coordinates

### 1. A plane through the origin—or a shifted plane?

Question. In $\mathbb{R}^3$, compare $S = \{(a, b, 0)^T : a, b \in \mathbb{R}\}$ and $T = \{(a, b, 1)^T : a, b \in \mathbb{R}\}$. Which is a subspace? Give its basis and dimension, and locate $(2, 3, 1)^T$.

Solution. S contains zero. Adding two of its vectors keeps the third coordinate zero, and multiplying by any real scalar also keeps it zero. Thus S is a subspace. Every vector in S equals $a(1, 0, 0)^T + b(0, 1, 0)^T$. These two spanning vectors are independent, so they form a basis and $\dim S = 2$. T excludes zero: even zero times a vector of T leaves T. It is not a subspace. The vector $(2, 3, 1)^T$ belongs to T but not to S.

### 2. A different basis, the same vector

Question. Use $b_1 = (1, 1)^T$ and $b_2 = (1, -1)^T$ to represent $v = (6, 2)^T$. Establish that the representation is unique.

$$c_1 + c_2 = 6, \qquad c_1 - c_2 = 2 \qquad \Rightarrow \qquad c_1 = 4,\ c_2 = 2$$

Solution. Adding the equations gives $2c_1 = 8$; subtraction gives $2c_2 = 4$. Reconstruction yields $4(1, 1)^T + 2(1, -1)^T = (6, 2)^T$. The basis matrix has determinant $-2$, which is nonzero, so its columns are independent and the coordinates $(4, 2)^T$ are unique. They are coefficients in this basis, not a new position for v.

# Level B · Rank and fundamental subspaces

## 3. Describe all three spaces of a rectangular matrix

Question. For A = [(1, 2, 3); (2, 4, 6)], find bases for its column, row and null spaces. Check rank–nullity and identify their ambient spaces.

Solution. All columns are multiples of $(1, 2)^T$; the second row is twice the first. A column-space basis is $\{(1, 2)^T\}$ in $\mathbb{R}^2$, and a row-space basis is {(1, 2, 3)} in $\mathbb{R}^3$. Both spaces have dimension 1. The null-space equation reduces to $x_1 + 2x_2 + 3x_3 = 0$. Write $x_2 = s$ and $x_3 = t$.

$$\mathrm{x} = \mathrm{s}(-2,\, 1,\, 0)^{\mathrm{T}} + \mathrm{t}(-3,\, 0,\, 1)^{\mathrm{T}}$$

These two vectors lie in $\mathbb{R}^3$, are independent, and are both sent to zero by A. They form a null-space basis. Hence nullity is 2 and rank + nullity = 1 + 2 = 3, the number of columns. The row and null spaces are perpendicular: each null basis vector has dot product zero with (1, 2, 3). The column space is in a different ambient space and must not be directly compared to these three-coordinate spaces.

## 4. Redundant columns and nonunique coefficients

Question. Let B have columns $(1, 2)^T$ and $(2, 4)^T$. Represent $y = (3, 6)^T$. Are its coefficients unique? Does adding the second column increase the span?

$$\mathrm{c}_1 + 2\mathrm{c}_2 = 3 \quad \Rightarrow \quad \mathrm{c} = (3 - 2\mathrm{t},\, \mathrm{t})^{\mathrm{T}},\ \mathrm{t} \in \mathbb{R}$$

Solution. Every such pair reconstructs y. Examples are $(3, 0)^T$ and $(1, 1)^T$, so the coefficients are not unique. The second column is twice the first; it adds no new direction. Rank(B) = 1 and its span is the line through $(1, 2)^T$. This shows why a spanning collection is not necessarily a basis, and why unique fitted vectors do not imply unique coefficient vectors.

## Numerical verification

Use matrix multiplication to verify each null-space vector and reconstruction. If SVD returns different signed basis vectors, compare the spaces or their orthogonal projectors instead of expecting identical entries. For nearly dependent columns, record the rank tolerance.

# Level B · Constructing an orthogonal projector

## 5. Project onto a line and prove closest distance

Question. Project $x = (2, 3)^T$ onto span{u}, where $u = (1, 1)^T$. Find P, the projected vector p and residual r. Why is p the closest point?

$$P = uu^T/(u^Tu) = [(1/2,\ 1/2);\ (1/2,\ 1/2)]$$

$$p = Px = (5/2,\ 5/2)^T, \qquad r = (-1/2,\ 1/2)^T$$

Solution. Since $u^Tu = 2$ and $u^Tx = 5$, the coefficient is 5/2. Symmetry is immediate, and multiplying P by itself gives P. Also $u^Tr = 0$ and $p^Tr = 0$. For any z in the line, $p - z$ lies in the line and is perpendicular to r. Therefore $\|x - z\|^2 = \|r\|^2 + \|p - z\|^2$, minimized at $z = p$. Here $\|x\|^2 = 13$, $\|p\|^2 = 25/2$, and $\|r\|^2 = 1/2$.

## 6. Projection onto a plane and a partial plot

Question. Project $x = (2, 3, 4)^T$ onto $S = \mathrm{span}\{(1, 0, 0)^T, (0, 1, 0)^T\}$. Explain what is lost from a plot showing only coordinates 1 and 3.

$$P = \mathrm{diag}(1,\ 1,\ 0), \qquad p = (2,\ 3,\ 0)^T, \qquad r = (0,\ 0,\ 4)^T$$

Solution. P keeps the first two coordinates and removes the third. It is symmetric and idempotent. The residual is perpendicular to both spanning vectors. The squared-length check is 29 = 13 + 16. In the notebook’s two-coordinate view, p appears as (2, 0), but that is not the full vector. The omitted second coordinate remains 3. Reading the plotted arrow as the full projection would incorrectly discard retained information.

## Check to retain in every implementation

For an orthonormal basis Q, verify $P = QQ^T$, $P^T \approx P$, $P^2 \approx P$ and $Q^T(x - Px) \approx 0$. Reconstruction $x \approx p + r$ is necessary but not sufficient to prove orthogonality.

# Level C · Regression tests and least squares

## 7. A zero leading column must not change the answer

Question. For B = [(0, 0); (0, 1)], find the orthogonal projector onto Col(B). Explain why column order must not affect the projector.

Solution. The first column is zero and the second is $(0, 1)^T$, so the column space is the second coordinate axis. Its orthonormal basis can be $Q = (0, 1)^T$; therefore $P = QQ^T = \mathrm{diag}(0, 1)$. Swapping the columns does not change their span or the projector. Simply retaining the first rank(B) columns from an unpivoted QR result can select the wrong direction. An SVD-based column-space basis handles this case. A regression test should check the known projector and its invariance under column permutation, not only $P^2 = P$.

## 8. Fit a line and verify residual orthogonality

Question. Fit an intercept and slope to responses $b = (1, 2, 2)^T$ observed at t = 0, 1, 2. Find coefficients, fitted values, residual and squared error.

$$\mathrm{A} = [(1,\ 0);\ (1,\ 1);\ (1,\ 2)]$$

$$\mathrm{A}^{\mathrm{T}}\mathrm{A} = [(3,\ 3);\ (3,\ 5)], \qquad \mathrm{A}^{\mathrm{T}}\mathrm{b} = (5,\ 6)^{\mathrm{T}}$$

Solution. The normal equations are $3\beta_0 + 3\beta_1 = 5$ and $3\beta_0 + 5\beta_1 = 6$. Subtraction gives $2\beta_1 = 1$; then $\beta_0 = 7/6$. These equations provide a hand derivation; use a stable least-squares solver in numerical work rather than explicitly inverting $A^TA$.

$$\beta = (7/6,\ 1/2)^{\mathrm{T}}, \qquad \mathrm{A}\beta = (7/6,\ 5/3,\ 13/6)^{\mathrm{T}}$$

$$\mathrm{r} = (-1/6,\ 1/3,\ -1/6)^{\mathrm{T}}, \qquad \|\mathrm{r}\|^2 = 1/6$$

The residual sum is −1/6 + 1/3 − 1/6 = 0. Its inner product with the time column is 0 + 1/3 − 2/6 = 0. Hence $A^Tr = 0$, establishing that the fitted vector is the orthogonal projection onto Col(A). Independent design columns make β unique in this example.

# Level C · Interpreting mathematical guarantees

## 9. Is every idempotent matrix an orthogonal projector?

Question. Consider H = [(1, 1); (0, 0)]. Compute $H^2$ and $H^T$. For x = $(0, 1)^T$, inspect p = Hx and r = x − p.

$$H^2 = H, \qquad H^T = [(1,\,0);\,(1,\,0)] \neq H$$

$$p = (1,\,0)^T, \qquad r = (-1,\,1)^T, \qquad p^T r = -1$$

Solution. H is idempotent but not symmetric. Its residual is not perpendicular to its range, the first coordinate axis. H is an oblique projection, not the Euclidean orthogonal projection. The closest point on that axis to $(0, 1)^T$ is actually zero. This counterexample explains why an idempotence-only test is insufficient.

## 10. Unique fitted values, multiple coefficients

Question. Let A = [(1, 1); (1, 1)] and b = $(1, 3)^T$. Find the fitted vector and all least-squares coefficients. Which has minimum Euclidean norm?

Solution. The column space is span$\{(1, 1)^T\}$; the orthogonal projection of b is its coordinate mean times this vector, giving fitted values $(2, 2)^T$. The residual $(-1, 1)^T$ is perpendicular to that line. Every β satisfying $\beta_1 + \beta_2 = 2$ produces the same fitted values. Write β = $(1 + t, 1 - t)^T$. Its squared norm is $2 + 2t^2$, minimized at t = 0, so the minimum-norm solution is $(1, 1)^T$.

## Application distinction

PCA projects centered data onto selected orthonormal principal directions. A learned linear map in an attention architecture is not guaranteed to be a projector: it may be rectangular, nonsymmetric or nonidempotent. The label “projection” is not sufficient evidence of the orthogonal-projection properties used above. Specify the actual map and test the relevant assumptions.

# Capstone · A policy residual is not a verdict

## 11. Reproduce the illustrative policy decomposition

Question. Let $b_1 = (1, 0.2, 0.6)^T$, $b_2 = (0.2, 1, 0.6)^T$, $B = [b_1\ b_2]$, and $x = (0.9, 0.7, 0.1)^T$. Coordinates describe standardized Agriculture, Health and Energy priorities. Find the projection and interpret the residual. These are illustrative values, not empirical estimates.

$$\mathrm{B^TB} = [(7/5,\ 19/25);\ (19/25,\ 7/5)], \qquad \mathrm{B^Tx} = (11/10,\ 47/50)^\mathrm{T}$$

$$\mathrm{c} = (43/72,\ 25/72)^\mathrm{T}, \qquad \mathrm{p} = \mathrm{Bc} = (2/3,\ 7/15,\ 17/30)^\mathrm{T}$$

$$\mathrm{r} = (7/30,\ 7/30,\ -7/15)^\mathrm{T}$$

Solution. Solve the two coefficient equations to obtain c above. Multiplication by B reconstructs p; subtracting it from x yields r. Both archetype dot products with r are zero: for $b_1$, $7/30 + 7/150 - 7/25 = 0$, and $b_2$ gives the same sum with its first two terms exchanged. The negative Energy residual says that the proposal is lower on that coordinate than its fitted component. It does not prove inadequacy or establish a spending recommendation.

## 12. Write a defensible interpretation

Question. What should a decision-maker be told before using this fit? Would this projection automatically respect nonnegative coefficients summing to one?

Solution. State how each coordinate was standardized, why the two archetypes were chosen, and why Euclidean distance is appropriate. Report both p and r, verify rank and orthogonality, and examine sensitivity to plausible changes in scaling or archetypes. No: a linear subspace allows arbitrary real coefficients. Here the coefficients sum to 17/18, not one. If weights must be nonnegative and sum to one, solve an explicitly constrained problem. Even a mathematically correct projection is conditional on the representation and cannot substitute for evidence, feasibility checks or accountable judgment.

Reference: owner-approved Chapter 5, sections 5.1–5.6, and companion notebook v0.2.2. New exercises 9–10 are explicit counterexamples/extensions of those definitions. Verify calculations with the packaged mathematical QA script.